

Proof. **Step1. Degenerate and boundary cases.**

(a) $n = 0$. $(1 + x)^0 = 1$ and $1 + 0 \cdot x = 1$; equality holds for every admissible x .

(b) $x = 0$. $(1 + 0)^n = 1$ and $1 + n \cdot 0 = 1$; equality holds for all n .

(c) $x = -1$.

If $n = 0$ we are back to case (a).

If $n \geq 1$, $(1 + (-1))^n = 0$, $1 + n(-1) = 1 - n$,

$$0 \geq 1 - n \iff n \geq 1,$$

which is true by hypothesis. Equality $0 = 1 - n$ forces $n = 1$.

Henceforth we assume

$$n \geq 1, \quad x > -1 \quad (\text{so } 1 + x > 0).$$

Step2. Induction on the exponent n .

We prove the inequality for all integers $n \geq 0$ by induction.

Base step. $n = 0$ has been settled in Step1(a).

Induction hypothesis. Fix a real $x \geq -1$. Assume that for some integer $k \geq 0$

$$(1 + x)^k \geq 1 + kx. \tag{IH}$$

Our goal is to show

$$(1 + x)^{k+1} \geq 1 + (k + 1)x. \tag{Goal}$$

Step3. Inductive step (algebraic manipulation).

$$(1 + x)^{k+1} = (1 + x)^k(1 + x).$$

Since $1 + x \geq 0$ (by $x > -1$), multiplying the inequality (IH) by the non-negative factor $1 + x$ preserves the order:

$$(1 + x)^{k+1} \geq (1 + kx)(1 + x). \tag{1}$$

Expanding the right-hand side,

$$(1 + kx)(1 + x) = 1 + (k + 1)x + kx^2. \tag{2}$$

Substituting (2) into (1) gives

$$(1 + x)^{k+1} \geq 1 + (k + 1)x + kx^2. \tag{3}$$

Step4. Non-negativity of the extra term.

Because $k \geq 0$ and $x^2 \geq 0$ for every real x , we have $kx^2 \geq 0$. Hence

$$1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x,$$

and together with (3) we obtain the desired inequality (Goal):

$$(1 + x)^{k+1} \geq 1 + (k + 1)x.$$

Thus the induction step is complete, and by the principle of mathematical induction the inequality

$$(1+x)^n \geq 1+nx$$

holds for all integers $n \geq 0$ and all real $x \geq -1$.

Step5. Alternative proof via the binomial theorem (optional).

The binomial theorem yields

$$(1+x)^n = \sum_{i=0}^n \binom{n}{i} x^i = 1 + nx + \sum_{i=2}^n \binom{n}{i} x^i.$$

Factor x^2 :

$$\sum_{i=2}^n \binom{n}{i} x^i = x^2 \sum_{j=0}^{n-2} \binom{n}{j+2} x^j, \quad (j = i - 2).$$

All binomial coefficients $\binom{n}{j+2}$ are non-negative, and for $x \geq 0$ each $x^j \geq 0$; for $-1 \leq x < 0$ we have $|x| \leq 1$ and consequently the sum remains non-negative. Therefore

$$(1+x)^n = 1 + nx + \underbrace{x^2 \left(\sum_{j=0}^{n-2} \binom{n}{j+2} x^j \right)}_{\geq 0} \geq 1 + nx.$$

Step6. Equality conditions.

From (3) we see that equality

$$(1+x)^{k+1} = 1 + (k+1)x + kx^2$$

holds precisely when the extra term kx^2 vanishes, i.e. when $k = 0$ or $x = 0$.

- $k = 0$ corresponds to the base case $n = 0$; then $(1+x)^0 = 1 = 1 + 0 \cdot x$ for any $x \geq -1$.
- $x = 0$ gives $(1+0)^n = 1 = 1 + n \cdot 0$ for every $n \geq 0$.
- The boundary case $x = -1$ was treated in Step1(c); equality occurs only for $n = 1$ (or the trivial $n = 0$ case).

Collecting all possibilities,

$\begin{aligned} n &= 0 && (\text{any } x \geq -1), \\ x &= 0 && (\text{any } n \geq 0), \\ x &= -1 \text{ and } n = 1. \end{aligned}$
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